

Homework 2. Markov decision problems

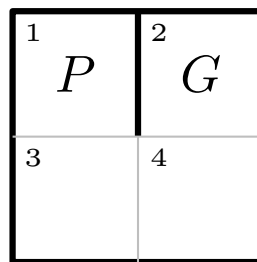


Figure 1: A taxi driver must pick up a passenger standing in the cell marked with “P” and drop it in the cell marked with “G”.

Consider a taxi driver moving in the grid-world environment of Fig. 1. The driver must pick up a passenger (the passenger is in the cell marked with “P”) and drop her at the cell marked with “G”.

At each step, the agent may move in any of the four directions—up, down, left and right. Movement across a *grey* cell division succeeds with a 0.8 probability and fails with a 0.2 probability. Movements across black cell divisions are not possible. When the movement fails, the driver remains in the same cell.

To pick up the passenger, the taxi driver simply needs to stand in the corresponding cell. In other words, as soon as the taxi driver stands on the cell marked with “P”, you may consider that the passenger is in the car. For simplicity, assume that once the passenger is in the taxi, it remains there forever—the taxi driver should, therefore, go to “G” and not leave.

Exercise 1.

- Identify the state space, $\mathcal{X}$, and the action space, $\mathcal{A}$, for the MDP.
- Write down the transition probability matrix for the action “down” and a (possible) cost function for the MDP. Make sure that the cost function is as simple as possible and verifies

$c(x, a) \in [0, 1]$ for all states $x \in \mathcal{X}$ and actions $a \in \mathcal{A}$. Note, in particular, that the cost should depend only on the taxi driver completing the delivery of the passenger.

- (c) Compute the cost-to-go function associated with the policy in which the taxi driver always goes down, using a discount $\gamma = 0.9$. You can use any software of your liking for the harder computations, but should indicate all other computations.

Solution:

- (a) We note, first of all, that the actions of the taxi driver depend on whether the passenger is on the taxi or not. If it is, then the taxi driver should head to the goal. If not, it should head to the cell marked with “P”. Therefore, the state should include information about both the position of the taxi and whether or not the passenger is onboard. This leads to the state space

$$\mathcal{X} = \{2_\emptyset, 3_\emptyset, 4_\emptyset, 1_p, 2_p, 3_p, 4_p\},$$

where we did not include $1_\emptyset$ since we assume that as soon as the taxi stands on cell 1, the passenger immediately gets on the car. The action space is

$$\mathcal{A} = \{U, D, L, R\},$$

each representing a direction for the movement of the taxi.

- (b) The transition probability for the action “down” (D) is given by

$$\mathbf{P}_D = \begin{bmatrix} 0.2 & 0.0 & 0.8 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 1.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 1.0 & 0.0 & 0.0 & 0.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.2 & 0.0 & 0.8 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.2 & 0.0 & 0.8 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 1.0 & 0.0 \\ 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 0.0 & 1.0 \end{bmatrix}.$$

A possible cost function penalizes the taxi driver for not being at the goal (which is

state 2_p), yielding

$$\mathbf{C} = \begin{bmatrix} 1.0 & 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 & 1.0 \\ 0.0 & 0.0 & 0.0 & 0.0 \\ 1.0 & 1.0 & 1.0 & 1.0 \\ 1.0 & 1.0 & 1.0 & 1.0 \end{bmatrix}.$$

- (c) The cost-to-go can easily be computed by noting the following: from all states except 2_p the taxi driver will incur a cost of 1 at every single time-step when following the policy that always goes down. On the other hand, when departing from state 2_p , the taxi driver will incur an initial cost of 0, and of 1 in every subsequent time-step. Therefore, for all initial states $x \neq 2_p$,

$$J(x) = \frac{1}{1 - \gamma} = 10.$$

On the other hand, for state 2_p ,

$$J(2_p) = 0 + \gamma(0.8 \times 10 + 0.2 \times J(2_p)).$$

Solving for $J(2_p)$ yields

$$J(2_p) = 8.78.$$